

What is the OSI Model?

Definition

The **OSI (Open Systems Interconnection) Model** is a 7-layer reference model used to understand and standardize how data is transmitted from one computer to another over a network. Each layer has a specific responsibility, making communication easier to design, troubleshoot, and understand.

Why Do We Need the OSI Model?

Without the OSI Model, network communication would be difficult to understand and troubleshoot.

The OSI Model divides the communication process into **seven independent layers**, where each layer performs a specific task.

Benefits

- Standardizes network communication
 - Simplifies troubleshooting
 - Makes networking easier to understand
 - Ensures interoperability between different vendors
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OSI Model Layers

Layer Name	Responsibility
7	Application Provides network services to end-user applications
6	Presentation Translation, Encryption, Compression
5	Session Establishes, manages and terminates sessions
4	Transport End-to-end communication using TCP/UDP
3	Network IP Addressing and Routing

Layer Name	Responsibility
2 Data Link	MAC Addressing and Local Delivery
1 Physical	Transmission of Bits

Mnemonic

A Pizza Shop Takes New Delivery Packages

Application
Presentation
Session
Transport
Network
Data Link
Physical

1. Application Layer (Layer 7)

What does it do?

The **Application Layer** is the layer closest to the user. It provides network services to applications so users can communicate over a network.

Example

You open **Google Chrome** and visit **google.com**.

The Application Layer receives your request and starts the communication.

Common Protocols

- HTTP
 - HTTPS
 - FTP
 - DNS
 - SMTP
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2. Presentation Layer (Layer 6)

What does it do?

The **Presentation Layer** prepares the data before sending it.

Its main responsibilities are:

- Data Translation
- Encryption / Decryption
- Compression

Example

When you open an **HTTPS** website, this layer encrypts your data before it is transmitted over the network, ensuring secure communication.

3. Session Layer (Layer 5)

What does it do?

The **Session Layer** establishes, manages, and terminates communication sessions between devices.

Example

When you join a **Zoom** or **Microsoft Teams** meeting:

- It creates the session.
 - It keeps the session active while the meeting is running.
 - It closes the session when the meeting ends.
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4. Transport Layer (Layer 4)

What does it do?

The **Transport Layer** is responsible for end-to-end data delivery.

Its responsibilities include:

- Reliable communication (**TCP**)
- Fast communication (**UDP**)

- Error checking
- Data segmentation

Example

When downloading a file, **TCP** ensures that every part of the file arrives correctly and in the correct order.

5. Network Layer (Layer 3)

What does it do?

The **Network Layer** determines the best path for data to reach its destination.

It uses:

- IP Address
- Routers

Example

When your laptop sends data to a server located in another country, the Network Layer selects the most efficient route for the data.

6. Data Link Layer (Layer 2)

What does it do?

The **Data Link Layer** provides communication between devices within the same local network.

It uses:

- MAC Address
- Switches

Example

Your laptop sends data to your Wi-Fi router using MAC addresses.

7. Physical Layer (Layer 1)

What does it do?

The **Physical Layer** transmits the actual bits over the physical medium.

Examples of Physical Media

- Ethernet Cable
- Fiber Optic Cable
- Wi-Fi Radio Signals

Example

The binary data (**0s and 1s**) is transmitted as electrical, optical, or wireless signals.

One-Line Purpose of Each Layer (Quick Revision)

Layer	Purpose
Application	Provides network services to end-user applications.
Presentation	Translates, encrypts/decrypts, and compresses data.
Session	Establishes, manages, and terminates communication sessions.
Transport	Ensures reliable or fast data delivery (TCP/UDP).
Network	Handles logical addressing and routing (IP).
Data Link	Uses MAC addresses and handles communication within the local network.
Physical	Transmits data as electrical, optical, or wireless signals.